

Abhinav Rajan, PhD

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[Google Scholar](#) | [Web of Science](#) | [LinkedIn](#)

PROFILE

Postdoctoral Fellow in Mechanical and Aeronautical Engineering at the University of Pretoria, specialising in solar energy systems. Nine journal publications, 150+ citations, an h-index of 6, and three years of undergraduate teaching experience in thermodynamics and heat transfer. Completed a PhD in Mechanical Engineering at IIT Madras (2024) and served as a Research Associate at IIT Patna, working on thermal-fluid systems and payload design for stratospheric applications. Research covers solar thermal systems, thermal energy storage, thermal management, CFD, and machine learning, with results published in journals including Energy Conversion and Management, Renewable Energy, and Energy. Collaborates with researchers across the IITs, CSIR-CMERI, the University of Sydney, and the University of Hertfordshire.

EDUCATION

Doctor of Philosophy (PhD) – Mechanical Engineering

2019 – 2024

[Indian Institute of Technology Madras, Chennai, India](#) | GPA: 8.22/10

Supervisor: Prof K. S. Reddy, Department of Mechanical Engineering

Thesis: Investigations of Solar Parabolic Dish Collector with Thermal Energy Storage for Process Heating Applications

Master of Technology (MTech) – Thermal Engineering

2013 – 2015

[Maulana Azad National Institute of Technology, Bhopal, India](#) | GPA: 8.6/10

Supervisor: Prof K. R. Aharwal, Department of Mechanical Engineering

Thesis: Performance Analysis of Cold Storage for the Different Stacking Arrangements

Bachelor of Engineering (B.E.) – Mechanical Engineering

2009 – 2013

[Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal, India](#) | 79.72%

RESEARCH EXPERIENCE

Postdoctoral Fellow

Sep 2025 – Present

[University of Pretoria, Hatfield, South Africa](#)

PI: Prof Willem le Roux | Department of Mechanical & Aeronautical Engineering

- Developing an integrated structural-optical-thermal framework and training deep learning models to predict multifaceted dish system performance for high-temperature applications, including a community-based aluminium melting process for Maisa Village, South Africa.
- Collaborating with researchers at the University of Hertfordshire, UK, on a thermal storage system, through the International Science Partnerships Fund (ISPF).
- Presented findings at SASEC 2026 (South Africa) and accepted for a poster presentation at SolarPACES 2026 (Germany); manuscripts in preparation.

Tools: SolTrace, Python, SpaceClaim, ANSYS Fluent, MATLAB, Deep Learning (TensorFlow/Keras)

Research Associate-I

Aug 2024 – Aug 2025

[IIT Patna, Bihar, India](#)

PI: Dr Rishi Raj & Dr Ashwani Assam | Department of Mechanical Engineering

Project: IRDE/DRDO-sponsored – Development of Thermally Controlled Modules of Optronics Payloads for Stratospheric Operation

- Performed thermal-fluid simulations for stratospheric environments using ANSYS Fluent and Simcenter FLOEFD.
- Developed thermal control strategies to maintain payload temperatures within safe operational limits under stratospheric conditions.
- Conducted lab-scale experimental validation and data acquisition for prescribed stratospheric operating conditions.
- Prepared technical reports and collaborated with IRDE (DRDO) at Dehradun (India) on system integration and performance assessment.
- Presented poster at International Conference on Electro-Optic & Photonics (EOP-2025), IRDE, Dehradun, Nov 2025.

Tools: SolidWorks, SpaceClaim, ANSYS Fluent, Simcenter FLOEFD

Doctoral Researcher (PhD)

Jan 2019 – Jul 2024

- Developed an integrated optical-thermal model for solar parabolic dish collectors; validated against lab-scale experiments at the Heat Transfer and Thermal Power Laboratory. Applied deep learning (ANN, TensorFlow/Keras) to predict solar collector performance
- Investigated PCM-based thermal energy storage under varying geometric and operational parameters; performed large-scale simulations on the HPCE cluster.
- Contributed to the SUNSHINE Ladakh project (DST-funded, IIT Madras – IIT Delhi): designed solar cavity receivers for a 40 m² solar dish and thermal energy storage for drying, cooking, and space-heating applications, targeted for a residential school in the Ladakh region of India.
- Assisted master's student (published in Thermal Science and Engineering Progress, IF 5.4).
- Teaching Assistant for four graduate-level courses, held concurrently with doctoral research; Laboratory Demonstrator for undergraduate students and visitors (2022–2024).
- Published seven journal articles and presented at six international conferences.

Tools: SolTrace, ASAP, SolidWorks, ANSYS Fluent, COMSOL Multiphysics, MATLAB, Python, TecPlot, HPCE

Visiting Researcher

Jun 2023 – Aug 2023

University of Sydney, NSW, Australia

Host: Dr Yixiang Gan | School of Civil Engineering

- Developed a numerical model for PCM-based thermal energy storage for low-temperature process heating; investigated charging dynamics for effective energy storage capacity.
- Funded by the IIE Award (Office of Global Engagement, IIT Madras) and Alumni Travel Grant, IIT Madras.
- Results published in Journal of Energy Storage (IF 9.8, 2024).

Tools: SolidWorks, ANSYS Fluent

Postgraduate Researcher (MTech)

Jul 2013 – Jun 2015

Maulana Azad National Institute of Technology, Bhopal

Supervisor: Prof K. R. Aharwal | Department of Mechanical Engineering

- Experimentally investigated 3 kW cold storage using a Vapour Compression Refrigeration System.
- Analysed the effect of stacking arrangement (10 cm, 20 cm, 30 cm gaps) on cold storage performance.

TEACHING EXPERIENCE

Assistant Professor

Sep 2015 – Sep 2018

Netaji Subhas Institute of Technology, Patna

- Taught undergraduate courses in Thermodynamics, Heat Transfer, and Refrigeration & Air Conditioning over a three-year appointment.
- Supervised student projects and provided academic mentoring to undergraduates.

Teaching Assistantship – IIT Madras (under Prof K. S. Reddy)

- ME 6148 Renewable Energy Technology – Jan–May 2023
- ME 5129 Principles of Thermal Energy Conversion – Jul–Nov 2022
- ME 6127 Energy and Environment – Jan–May 2022
- ME 6005 Solar Energy for Process Heat & Power Generation – Jul–Nov 2021

JOURNAL PUBLICATIONS (9)

[J1] **Rajan, A.** & Reddy, K. S. (2025). Integrated optical-thermal model to predict the performance of a solar parabolic dish collector for process heating applications. Energy, 321, 135176. <https://doi.org/10.1016/j.energy.2025.135176> [IF: 9.4]

[J2] Kumar, K. H., **Rajan, A.**, Karmakar, M., Mondal, B., Mallisetty, P. K. & Reddy, K. S. (2025). Comparative thermal assessment of Al₂O₃/EG and ZnO/EG as heat transfer fluids in solar parabolic dish receiver. Renewable Energy, 249, 123216. <https://doi.org/10.1016/j.renene.2025.123216> [IF: 9.1]

[J3] Abhijit Rajan, Prasad, B. N., Kumar, R., Dhuria, G. K., Kumar, A. & **Rajan, A.** (2025). An experimental approach to heat transfer enhancement in heat exchanger tubes using hyperbolic-cut twisted tape inserts: IoT and biomedical insight. Multiscale & Multidisciplinary Modelling, Experiments and Design, 8(9), 384. <https://doi.org/10.1007/s41939-025-00951-8> [IF: 2.0]

[J4] **Rajan, A.**, Gan, Y. & Reddy, K. S. (2024). Performance assessment of phase change material-based thermal energy storage. Journal of Energy Storage, 87, 111424. <https://doi.org/10.1016/j.est.2024.111424> [IF: 9.8]

- [J5] Rajan, A. & Reddy, K. S. (2024).** Integrated optical-thermal model and deep learning technique to estimate the performance of a conical cavity receiver coupled solar parabolic dish collector. *Energy Conversion and Management*, 301, 118052. <https://doi.org/10.1016/j.enconman.2023.118052> [IF: 10.9]
- [J6] Mandal, P., Rajan, A. & Reddy, K. S. (2024).** Optical and thermal investigation of hyperbolic cavity receiver with secondary reflector for solar parabolic dish collector. *Thermal Science and Engineering Progress*, 47, 102350. <https://doi.org/10.1016/j.tsep.2023.102350> [IF: 5.4]
- [J7] Rajan, A. & Reddy, K. S. (2023).** Integrated optical and thermal model to investigate the performance of a solar parabolic dish collector coupled with a cavity receiver. *Renewable Energy*, 219, 119376. <https://doi.org/10.1016/j.renene.2023.119376> [IF: 9.1]
- [J8] Rajan, A. & Reddy, K. S. (2023).** Convective heat loss prediction from conical cavity receiver of solar parabolic dish collector using numerical method and artificial neural network. *Numerical Heat Transfer, Part A: Applications*, 83(6), 626–649. <https://doi.org/10.1080/10407782.2022.2102338> [IF: 3.5]
- [J9] Rajan, A. & Reddy, K. S. (2022).** Optical modeling of corrugation cavity receiver for large-aperture solar parabolic dish collector. *Energy Sources, Part A: Recovery, Utilisation, and Environmental Effects*, 44(2), 3330–3348. <https://doi.org/10.1080/15567036.2022.2063458> [IF: 2.2]

CONFERENCE PROCEEDINGS (8)

- [C1] Rajan, A., le Roux, W. G., Humphries, E. D., Craig, K. J., Cox, R. B. & van Vuuren, D. J. (2026, Sept 15–18).** Modelling the effect of tracking error on multifaceted dish system performance for high-temperature applications. 32nd SolarPACES Conference, Bad Neuenahr, Germany. (Accepted)
- [C2] Rajan, A., Cox, R. B., le Roux, W. G., van Vuuren, D. J. & Craig, K. J. (2026, Jun 24–26).** Analysing the impact of tracking error on an air-heating open-cavity tubular receiver in a solar parabolic dish system. Southern African Sustainable Energy Conference (SASEC 2026), Kruger National Park, SA.
- [C3] Rajan, A. & Reddy, K. S. (2024, Aug 16).** Deep learning model to predict the performance of solar parabolic dish collector with conical cavity receiver. 1st International Symposium on Renewable Energy Technologies: Energising the Future, IIT Delhi, India.
- [C4] Rajan, A. & Reddy, K. S. (2024, Aug 12–14).** Optical and thermal analysis of solar parabolic dish collector with hetero-conical cavity receiver for medium temperature applications. 21st International Conference on Sustainable Energy Technologies (SET 2024), Shanghai, China.
- [C5] Mondal, P., Rajan, A. & Reddy, K. S. (2023, Oct 30–Nov 4).** Thermo-optical modeling for hyperbolic cavity receiver of 40 m² parabolic dish collector. ISES Solar World Congress 2023, New Delhi, India.
- [C6] Rajan, A. & Reddy, K. S. (2023, Aug 14–18).** Coupled optical-thermal modeling of solar parabolic dish collector for novel cavity receiver. 17th International Heat Transfer Conference (IHTC-17), Cape Town, South Africa.
- [C7] Rajan, A. & Reddy, K. S. (2023, May 18–20).** Comparative study of cylindrical and triple concentric tube models for PCM-based thermal energy storage. 11th European Conference on Renewable Energy Systems (ECRES 2023), Riga, Latvia.
- [C8] Rajan, A. & Reddy, K. S. (2021, Sept 22–23).** Estimation of convective heat losses from conical cavity receiver of solar parabolic dish collector under wind conditions and receiver orientations. 19th International Stirling Engine Conference (ISEC 2021), Online.

PRESENTATIONS & POSTERS (2)

- [P1] Rajan, A., Sharma, A. K., Mishra, M. P., Ashwani, A. & Raj, R. (2025, Nov 19).** Enabling reliable optronic operations in extreme stratospheric conditions. International Conference on Electro-Optic & Photonics (EOP-2025), IRDE, Dehradun, India.
- [P2] Rajan, A. & Reddy, K. S. (2021, Dec 27–29).** Optical modeling of solar parabolic dish collector coupled with conical cavity receiver. 6th International Conference on Sustainable Energy and Environmental Challenges, Lucknow, India.

BOOK CHAPTERS (1)

- [B1] Rajan, A., Goel, T., Rakshit, D. (2026),** Strategies for Integrating Artificial Intelligence and IoT to Enhance the Performance of Clean Energy Infrastructure, Springer (under Review)

AWARDS & FUNDING

- Postdoctoral Grant Award, University of Pretoria, South Africa (2025)

- Alumni Travel Grant, IIT Madras, India (2024)
- International Immersion Experience (IIE) Award, Office of Global Engagement, IIT Madras (2023)
- Conference Travel Grant – paper presentation at University of Latvia, Riga (2023)
- Half-Time Research Assistantship for PhD, funded by MHRD, Government of India (2019–2023)
- GATE Scholarship for MTech, funded by MHRD, Government of India (2013–2015)
- Qualified GATE (National Level Exam) – Mechanical Engineering (2013)
- Certificate of Merit – 3rd Semester, Sagar Institute of Research & Technology, Bhopal (2012)
- 10th Rank – Intermediate Examination, Bihar School Examination Board (2009)
- Hindustan Pratibha Samman, organised by HT Media (2009)

TECHNICAL SKILLS & TOOLS

CFD & Simulation: ANSYS Fluent (Expert), ANSYS Mechanical, COMSOL Multiphysics, Simcenter FLOEFD

Solar / Optical Tools: SolTrace (Expert), ASAP Ray Tracing Tool

CAD & Geometry: SolidWorks (Advanced), SpaceClaim (Advanced), TecPlot

Programming & ML: Python (Deep Learning, TensorFlow/Keras), MATLAB

Documentation & Publishing: LaTeX, Microsoft Office, WordPress

Media & Outreach: Filmora, CapCut

PROFESSIONAL MEMBERSHIPS & SERVICE

- Life Member, Indian Society for Heat and Mass Transfer (ISHMT) (2024–present)
- Peer Reviewer – Journal of Energy Storage (Elsevier), Frontiers in Mechanical Engineering
- Technical Reviewer – Southern African Sustainable Energy Conference (SASEC 2026), SolarPACES 2026
- Chaired a session, Track 6B Solar Thermal Energy and Thermal Storage in Southern African Sustainable Energy Conference (SASEC 2026), June 25, 2026.
- Organising Committee Volunteer – 2nd International Symposium on Renewable Energy Technologies: Energising the Future, IIT Madras, Aug 28, 2024

RESEARCH OUTREACH & MENTORING

- Assisted master's student at IIT Madras working on cavity receivers (published in Thermal Science and Engineering Progress, IF 5.4) and compound parabolic collectors.
- Mentoring Credence CleanTech Solutions, an India-based startup, on energy system design.
- Laboratory Demonstrator to Undergraduate Students and Visitors, IIT Madras (2022–2024).
- Student Coordinator – Technical Poster Making event, TECHNOSAGA 2011, during B.E. studies.

REFEREES

Key supervisors and collaborators include:

- Prof Willem le Roux, University of Pretoria, South Africa (Postdoc PI), willem.leroux@up.ac.za
- Prof K. S. Reddy, IIT Madras, India (PhD Supervisor), ksreddy@iitm.ac.in
- Prof (Rtd.) B.N. Prasad, NIT Jamshedpur, India (Mentor), bnprasad.nit@gmail.com
- Dr Rishi Raj - IIT Patna, India (Research Associate PI), rroj@iitp.ac.in
- Dr Ashwani Assam- IIT Patna, India (Research Associate PI), aashwani@iitp.ac.in
- Dr Vikas Kumar, MANIT Bhopal, India (Mentor), vikashism2012@gmail.com